The 1/3–2/3 Conjecture for Width-3 Posets:
A Two-Page Summary

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Statement. Let P be a finite poset (not a chain) and let $\mathcal{L}(P)$ be the set of its linear extensions under the uniform measure. A pair (x, y) of incomparable elements is *balanced* if $\mathbb{P}[x <_L y] \in [1/3, 2/3]$ for $L \sim \mathrm{Unif}(\mathcal{L}(P))$.

Conjecture (Kislitsyn–Fredman–Linial). *Every non-chain finite poset has a balanced pair.*

Open since 1968. The current best unconditional constant is $0.2764\ldots$ (Kahn–Linial 1991, refining Brightwell). It is known for width-2 posets (Linial 1984), 2-dimensional posets (Kahn–Saks 1984), semiorders (Brightwell), and symmetric posets, but *not* previously known at width 3.

Theorem 1 (Target result — see status note)**.** *Let P be a finite poset of width at most 3 that is not a chain. Then P admits a pair (x, y) of incomparable elements with $1/3 \le \mathbb{P}[x <_L y] \le 2/3$.*

Status. This is the result the paper aims at, and the full argument for it is given in `main.tex`. It is *not* at present a complete proof: it is conditional on an arithmetic-richness hypothesis (Hypothesis A of Step 8), and an internal audit (`docs/onethird-math-finished-audit.md`) has identified several load-bearing steps whose proofs are currently incomplete — in Step 2 (per-fiber stability), Step 4 (two-interface incompatibility), and the deep irreducible case of the Step 8 layered balanced-pair lemma. The summary below describes the intended strategy; the gaps are marked **[GAP:]** in the main text.

The two ideas

The proof rests on two complementary observations. The first is a *global* tool that turns a question about one number (the balance constant) into a question about the geometry of an exponentially large graph. The second is a *local* constraint — specific to width 3 — that prevents that geometry from accommodating an imbalanced poset.

Idea 1 — The balance constant is BK conductance. The Bubley–Karzanov graph $\mathrm{BK}(P)$ has vertex set $\mathcal{L}(P)$, with edges joining extensions that differ by a single P-compatible adjacent transposition. This is the natural local move on linear extensions; Bubley–Karzanov (1998) showed the corresponding random walk mixes in polynomial time.

The key reframing: suppose (x, y) is the most-balanced incomparable pair of an unbalanced P, with $\mathbb{P}[x < y] < 1/3$. Consider the cut $S_{xy} = \{L \in \mathcal{L}(P) : x <_L y\} \subseteq \mathrm{BK}(P)$. Then (i) S_{xy} has volume fraction $< 1/3$, and (ii) every boundary edge of S_{xy} swaps x and y specifically, which is rare since most extensions do not place x, y adjacent. So a counterexample produces a *low-conductance cut* in $\mathrm{BK}(P)$. A Cheeger-type inequality translates the original numerical statement ("$\delta(P) < 1/3$") into a geometric one. Instead of arguing about probabilities via correlation inequalities (Kahn–Saks, Brightwell, Kahn–Linial), we now have an entire graph and an entire cut to analyse, and the full strength of expansion methods becomes available.

Idea 2 — Width 3 makes the obstructions essentially 2D. What could prevent a low-conductance cut from existing? The local geometry of $\mathrm{BK}(P)$ near a generic "rich" incomparable pair (x, y) — one sharing many common neighbors — is a 2D grid: fixing the relative order of all elements outside $\{x, y\}$ and their joint neighborhood, the remaining freedom in L is recorded by two integer coordinates ("how far has x advanced?" / "how far has y advanced?") ranging over a region $D_{xy} \subseteq [0, t]^2$, and BK moves within this fiber are unit grid moves.

This 2D coordinate system exists *because* width ≤ 3. A fourth uninvolved element would push the local geometry to 3D, where many more boundary configurations become possible; at width 3 the only "other" element is one of the common neighbors, whose position is recorded inside the grid.

The 2D picture has two consequences. *(a) Local cuts are nearly monotone.* A standard edge-isoperimetric stability theorem says: a subset of a 2D grid region with small edge boundary is close to a monotone staircase. The restriction of any low-conductance cut to a rich fiber thus points in a definite direction — assigning each rich pair a canonical sign $\eta_{xy} \in \{+, -\}$ and axis. *(b) Two incoherent interfaces are incompatible.* If two rich pairs have monotone interfaces pointing in incompatible directions, their joint constraint on a shared fiber is over-determined: many BK boundary edges are forced. This converts "many rich pairs" into a violation of the conductance lower bound.

Putting it together. A width-3 counterexample P would force — by Idea 1 — a low-conductance cut S. By Idea 2(a), S is locally monotone on each rich fiber, giving each rich pair a canonical sign. By Idea 2(b), if the signs are *not* globally consistent, two incoherent rich pairs alone exceed the boundary budget. So the signs must be globally consistent — which forces P to admit a single linear potential function aligned with S, collapsing P to a layered, essentially 1-dimensional structure. But such posets are already balanced (Linial's width-2 result and a small-n enumeration cover the residual cases). Contradiction.

Why width 3 is essential. Both ingredients use width 3. The 2D grid coordinatisation requires that fixing two elements leaves no further incomparable third axis; at width ≥ 4 the local geometry is at least 3D and grid edge-isoperimetric stability is much weaker. The Dilworth dichotomy producing "enough rich interfaces" also relies on a decomposition into at most 3 chains. Beyond width 3, both fronts need genuinely new ingredients.

Repository. The full proof is in `main.tex` (assembling `step1.tex`–`step8.tex`); a partial Lean 4 formalisation is in `lean/`; a self-audit and reviewer pointers are in `summary.md` and `reviewers.md`. The README is candid about authorship, AI assistance, and the absence of external review — please read it before citing.